

Ashish Kumar Singh

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WORK EXPERIENCE

Bharat Electronics Limited (BEL)

August 2025 – Present

Software Development Engineer

New Delhi, IN

- Designed and developed backend services for **ICNS**, a mission-critical **Network Management System (NMS)** supporting **24×7 monitoring, configuration, telemetry ingestion, and operational control** across distributed communication infrastructure using **Java, Spring Boot, Hibernate/JPA, MySQL, PostgreSQL, and Maven**.
- Owned and delivered **10–12 backend modules** spanning device monitoring, alarm handling, reporting, RBAC, schedulers, audit logging, and operational workflows; built and enhanced **20+ REST APIs** for real-time monitoring, configuration management, and device lifecycle operations.
- Built an end-to-end **TCP/REST-based poller** for non-SNMP devices and developed backend workflows for **20–25+ monitoring and operational pages**, enabling centralized telemetry collection, live dashboards, traffic visualization, fault management, and remote-site monitoring across heterogeneous communication systems.
- Optimized **Hibernate/JPA mappings, SQL queries, caching, and asynchronous processing**, reducing average query latency by about **32%**, mitigating **N+1 query issues**, and improving performance for high-volume telemetry and operational data workloads.
- Modernized deployment of the NMS platform on **RHEL** by containerizing application, polling, analytics, trap-handling, and database services using **Podman, Quadlet, systemd, and shell automation**, reducing typical site deployment time from **2 weeks to 2–3 days** and lowering deployment effort from **2–3 engineers to 1**.
- Improved platform reliability and maintainability by contributing to **Pacemaker/Corosync-based HA failover, Spring Security RBAC, SSL/X.509 certificate workflows**, migration of legacy modules from **Struts to Spring Boot**, and resolution of **40+ production defects** across backend services and live deployment environments.

Defence Research and Development Organisation (DRDO)

June 2024 – August 2024

Software Developer Intern

New Delhi, IN

- Engineered an **eBPF-based TCP packet filtering system** at the NIC level, improving filtering efficiency by **37%**.
- Optimized **kernel-level packet processing** using embedded C, **libbpf**, and Linux APIs, achieving about **25% reduction in network latency** and increased throughput.
- Conducted performance benchmarking across multiple traffic patterns, validating scalability and stability under load.
- Collaborated with senior engineers to strengthen **network security and low-level system performance**.

TECHNICAL SKILLS

Programming Languages: Java, C, C++, Python, SQL, JavaScript

Backend & Web Technologies: Spring Boot, Spring Framework, Hibernate/JPA, MySQL, PostgreSQL, Node.js, Flask, REST APIs, React.js, HTML, CSS

Developer Tools & Platforms: Git, GitHub, Linux, Docker, Podman, systemd, Quadlet, VMware, Postman

Computer Science Fundamentals: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, Software Engineering

EDUCATION

National Institute of Technology, Jalandhar

Jalandhar, IN

Bachelor of Technology, Computer Science and Engineering

2021 - 2025

PROJECTS

RoleMate | *Java, Spring Boot, WebSocket, PostgreSQL, React, Docker*



- Built **RoleMate**, a real-time interview practice platform that matches users by **role, topic, and experience level**, enabling live text chat and browser-based video sessions using **Spring Boot, WebSockets, React, and PostgreSQL**.
- Designed a backend matchmaking engine with **role-based queues, scored partner matching, session lifecycle management, and per-role concurrency control** using **ReentrantLock**, refactoring monolithic logic into modular services for improved maintainability and scalability.
- Implemented **WebRTC signaling workflows** over WebSockets, including SDP offer/answer relay and **ICE candidate exchange**, allowing matched users to establish peer-to-peer audio/video sessions while keeping the backend as a lightweight signaling server.
- Added **asynchronous PostgreSQL persistence** for session metadata and user feedback using **Spring Data JDBC, Flyway, and @Async**, ensuring database writes did not block real-time matchmaking, chat, and signaling flows.
- Containerized the full stack with **Docker Compose, Nginx, PostgreSQL, and multi-stage builds**, and strengthened reliability with **55+ unit/integration tests** covering matchmaking, session handling and WebSocket workflows.